

Module 3: Polynomials and Their Shape – Prequiz

Name:

Date:

Four short questions. Use degree, leading term, roots, and multiplicity as appropriate.

1. Determine the degree and end behavior of

$$h(x) = -3x^6 + 2x^3 - 5.$$

2. For

$$p(x) = (x + 2)^3(x - 1)^2,$$

list the real roots and their multiplicities. State whether the graph crosses or touches the x -axis at each root.

3. What is the maximum possible number of turning points of a degree-5 polynomial?
4. Describe the end behavior of a polynomial with positive leading coefficient and odd degree.

Module 3: Polynomials and Their Shape – Weekly Check-In

Name:

Date:

Three questions. Use degree, leading term, roots, and multiplicity as appropriate.

1. Degree and end behavior.

For

$$f(x) = -2x^5 + 3x^2 - 7,$$

state the degree, identify the leading term, and describe the end behavior as $x \rightarrow -\infty$ and $x \rightarrow \infty$.

2. Roots and multiplicity.

For

$$p(x) = (x + 2)^2(x - 1)^3(x - 4),$$

list each real root and its multiplicity. At each root, state whether the graph crosses or touches the x -axis. What is the maximum possible number of turning points of p ?

Module 3: Polynomials and Their Shape – Weekly Check-In, continued

Name:

Date:

3. Sketching from factored form.

Consider

$$h(x) = -(x+3)(x+1)^2(x-2)^3.$$

Before graphing, state the degree and end behavior. Then sketch a possible graph of $h(x)$ below, showing the correct behavior at every real zero. Exact vertical scale is not required.

